

Statistical Regression

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TENTATIVE OUTLINE OF THE COURSE

- Simple Linear Regression
- Multiple Linear Regression
- Nonlinear Regression
- Generalised Linear Models
- Classification and Regression Trees
- *Random Forests*

BOOKS FOR PART 2

- Montgomery, D.C., Peck, E.A. and Vining G.G. (2012). Introduction to Linear Regression Analysis (Fifth Edition). Wiley.
- Chatterjee, S. and Hadi, A.S. (2012). Regression Analysis by Example (Fifth Edition). Wiley.

MULTIPLE LINEAR REGRESSION

- Consider again the `gestation` data and add other covariates to `Age` in forecasting `Weight`: first only `Births`, then only `SD` and finally both
- Do we improve R^2 ? Do we get meaningful results?
- $R^2 = 0.9668, 0.9763, 0.9319, 0.9464$ in the regressions below
- R^2 should not decrease adding a covariate: here the decrease, when introducing `SD`, is due to the reduced number of points since `SD` has many `NA`'s

```
library(GLMsData); data(gestation); attach(gestation)
reg=lm(Weight ~ Age)
summary(reg)
reg=lm(Weight ~ Age+Births)
summary(reg)
reg=lm(Weight ~ Age+SD)
summary(reg)
reg=lm(Weight ~ ., data=gestation)
summary(reg)
```

MULTIPLE LINEAR REGRESSION

- Multiple linear regression model: $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \varepsilon$
 - $\beta_0, \beta_1, \dots, \beta_p$ parameters referred as the model partial regression coefficients
 - Each β_i is said "partial" since measures the effect of each explanatory variable X_i on the response Y , given all the other explanatory variables
 - ε : random error component with $E(\varepsilon) = 0$ and $Var(\varepsilon) = \sigma^2$ unknown
 - Uncorrelated ε for different observations (y, x)

- Given n observations $(y_i, x_{i1}, \dots, x_{ip})$, $i = 1, \dots, n$, we get

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \varepsilon_i, \quad i = 1, \dots, n$$

- Simple linear regression is just a particular case with $p = 1$
- A second-order model **with interaction**, e.g.,

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_{11} X_1^2 + \beta_{22} X_2^2 + \beta_{12} X_1 X_2 + \varepsilon$$

becomes a multiple linear model with $X_1^2 = X_3$, $X_2^2 = X_4$, $X_1 X_2 = X_5$

MULTIPLE LINEAR REGRESSION

- Least squares method $\Rightarrow$ Minimise $S(\beta_0, \beta_1, \dots, \beta_p) = \sum_{i=1}^n \varepsilon_i^2 = \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2$
- $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$ least squares estimators (LSEs) of $\beta_0, \beta_1, \dots, \beta_p$ s.t.

$$\left. \frac{\partial S}{\partial \beta_0} \right|_{\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p} = -2 \sum_{i=1}^n \left(y_i - \hat{\beta}_0 - \sum_{j=1}^p \hat{\beta}_j x_{ij} \right) = 0$$

$$\left. \frac{\partial S}{\partial \beta_j} \right|_{\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p} = -2 \sum_{i=1}^n \left(y_i - \hat{\beta}_0 - \sum_{k=1}^p \hat{\beta}_k x_{ik} \right) x_{ij} = 0, \quad j = 1, \dots, p$$

- Simplifying, we get the least squares normal equations:

$$n\hat{\beta}_0 + \sum_{j=1}^p \left(\hat{\beta}_j \sum_{i=1}^n x_{ij} \right) = \sum_{i=1}^n y_i$$

$$\hat{\beta}_0 \sum_{i=1}^n x_{ij} + \sum_{k=1}^p \left(\hat{\beta}_k \sum_{i=1}^n x_{ij} x_{ik} \right) = \sum_{i=1}^n x_{ij} y_i, \quad j = 1, \dots, p$$

MULTIPLE LINEAR REGRESSION

$$n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^n x_{i1} + \hat{\beta}_2 \sum_{i=1}^n x_{i2} + \cdots + \hat{\beta}_k \sum_{i=1}^n x_{ik} = \sum_{i=1}^n y_i$$

$$\begin{array}{cccccc} \hat{\beta}_0 \sum_{i=1}^n x_{i1} & + & \hat{\beta}_1 \sum_{i=1}^n x_{i1}^2 & + & \hat{\beta}_2 \sum_{i=1}^n x_{i1}x_{i2} & + \cdots + \hat{\beta}_k \sum_{i=1}^n x_{i1}x_{ik} = \sum_{i=1}^n x_{i1}y_i \\ \vdots & & \vdots & & \vdots & & \vdots \end{array}$$

$$\hat{\beta}_0 \sum_{i=1}^n x_{ik} + \hat{\beta}_1 \sum_{i=1}^n x_{ik}x_{i1} + \hat{\beta}_2 \sum_{i=1}^n x_{ik}x_{i2} + \cdots + \hat{\beta}_k \sum_{i=1}^n x_{ik}^2 = \sum_{i=1}^n x_{ik}y_i$$

- Least squares normal equations
- I copied the page from the book: the authors use k instead of p
- We look for a more compact formulation!

MULTIPLE LINEAR REGRESSION

A more compact formulation is obtained using matrix notation

$$y = X\beta + \varepsilon$$

- y : $n \times 1$ vector of observations
- X : $n \times (p + 1)$ matrix of values of covariates
- β : $(p + 1) \times 1$: vector of regression coefficients
- ε : $n \times 1$ vector of random errors

$$y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \quad X = \begin{pmatrix} 1 & x_{11} & \vdots & x_{1p} \\ 1 & x_{21} & \vdots & x_{2p} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & \vdots & x_{np} \end{pmatrix}, \quad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}, \quad \varepsilon = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}$$

MULTIPLE LINEAR REGRESSION

- Find vector $\hat{\beta}$ of LSEs minimising

$$\begin{aligned} S(\hat{\beta}) &= \sum_{i=1}^n \varepsilon_i^2 = \varepsilon' \varepsilon = (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta) \\ &= \mathbf{y}'\mathbf{y} - \beta'\mathbf{X}'\mathbf{y} - \mathbf{y}'\mathbf{X}\beta + \beta'\mathbf{X}'\mathbf{X}\beta \\ &= \mathbf{y}'\mathbf{y} - 2\beta'\mathbf{X}'\mathbf{y} + \beta'\mathbf{X}'\mathbf{X}\beta \end{aligned}$$

since $\beta'\mathbf{X}'\mathbf{y}$ is a scalar $\Rightarrow$ its transpose $(\beta'\mathbf{X}'\mathbf{y})' = \mathbf{y}'\mathbf{X}\beta$

- The LSE should satisfy

$$\left. \frac{\partial S}{\partial \beta} \right|_{\hat{\beta}} = -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}\hat{\beta}$$

- $\Rightarrow$ least squares normal equations $\mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{X}'\mathbf{y}$
- Multiply both sides by $(\mathbf{X}'\mathbf{X})^{-1}$ (if it exists) $\Rightarrow \hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$
- $(\mathbf{X}'\mathbf{X})^{-1}$ exists if the covariates are linearly independent, i.e. no column of $\mathbf{X}$ is a linear combination of the other columns ($\Rightarrow n \geq p$, although, better, $n \gg p$)

MULTIPLE LINEAR REGRESSION

- Fitted regression model for $\mathbf{x}' = [1, x_1, \dots, x_p]$

$$\hat{y} = \mathbf{x}'\hat{\boldsymbol{\beta}} = \hat{\beta}_0 + \sum_{j=1}^p \hat{\beta}_j x_j$$

- Fitted values $\hat{y}_i$ corresponding to observed y_i

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \mathbf{H}\mathbf{y}$$

- $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$: Hat matrix $n \times n$ (mapping vector of observations into fitted one)
- $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{y} - \mathbf{H}\mathbf{y} = (\mathbf{I} - \mathbf{H})\mathbf{y}$: vector of residuals
- One can prove: $E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$ and $Var(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$

- σ^2 estimated by $SS_{Res} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n e_i^2 = \mathbf{e}'\mathbf{e} = \mathbf{y}'\mathbf{y} - \hat{\boldsymbol{\beta}}\mathbf{X}'\mathbf{y}$

- Unbiased estimator given by $\hat{\sigma}^2 = MS_{Res} = SS_{Res}/(n - p - 1)$

WEIGHTED LEAST SQUARES METHOD

- Earlier I mentioned the possibility of having $Var(\varepsilon_i) = \sigma^2/\omega_i \Rightarrow Var(y_i) = \sigma^2/\omega_i$, with ω_i a weight affecting the variability of the i -th observation
- Consider a $n \times n$ diagonal matrix $\mathbf{W}$ with ω_i 's on the diagonal
- Weighted Least Squares: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ and $Var(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{W}^{-1}$
- $\Rightarrow$ Minimise $S(\hat{\boldsymbol{\beta}}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'\mathbf{W}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \sum_{i=1}^n \omega_i (y_i - \mathbf{x}_i'\boldsymbol{\beta})^2$
- $\Rightarrow \hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}\mathbf{y}$
- The use of the weighted residual sum of squares (RSS) recognizes that some of the errors are more variable than others since cases with large values of ω_i will have small variance and will therefore be given more weight in the weighted RSS
- In Dunn and Smyth, the `gestation` dataset was considered with WLS (weighted LS) and not OLS (ordinary LS) since the data were average of n_i baby weights, each with variance σ^2 , so that their average had variance σ^2/n_i and $n_i = \omega_i$

CATEGORICAL VARIABLES

- So far we have considered either general covariates (without specifying their nature) or continuous one, but there are others which are categorical or qualitative (male/female, Europe/Africa/Asia/America/Oceania, bad/good)
- The response variable could be categorical, like success/failure in an exam (given number of hours of study) or repayment or not of a mortgage given age and salary of the borrower, but this will be the subject of future classes (e.g. logistic regression)
- We can consider dummy variables which are taking value either 0 or 1
 - If we are interested in the influence of the gender on, say, the relation between weight Y and height X_1 , we could introduce in the model a variable X_2 which takes value 1 if the person is a female and 0 if a male
 - The coefficient β_2 , if significant, will tell that, for the same height, the weight of a female will differ by β_2 from that of a male
- If there are L categories, more than 2 (e.g. 5 continents), then $L - 1$ dummy variables will be considered (e.g. X_2 for Africa, X_3 for Asia, X_4 for America and X_5 for Oceania) so that there will be a reference (Europe in this case) such that all the dummy variables are 0 for a European, while, e.g., only X_2 is 1 for an African and β_2 gives the difference in the response between an African and a European

TEST FOR SIGNIFICANCE OF REGRESSION

- This test determines if there is a linear relation between the response Y and the covariates $X_1, \dots, X_p$ and it is given by

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_p = 0 \text{ vs. } H_1 : \beta_j \neq 0 \text{ for at least one } j$$

- The comparison between the model without and with covariates is made considering an F_0 statistics with $(p, n - p - 1)$ degrees of freedom

$$F_0 = \frac{SS_{Reg}/p}{SS_{Res}/(n - p - 1)} \sim F_{p, n-p-1}$$

- $SS_{Res} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$ and $SS_{Reg} = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$

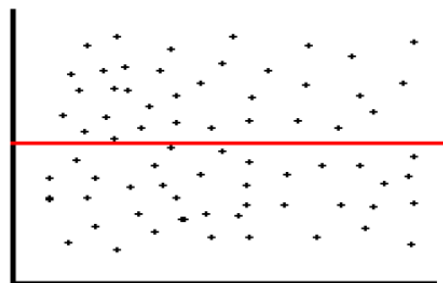
- Reject H_0 if $F_0 > F_{1-\alpha, p, n-p-1}$, quantile of order $1 - \alpha$ of the F distribution

MODEL CHECKING*

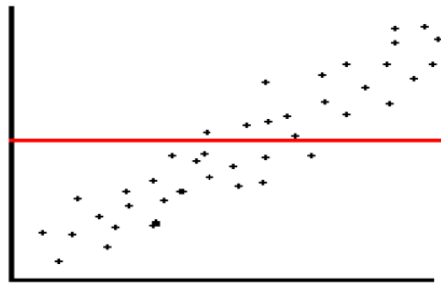
- *Linear relation*: A plot of the residuals e_i against the fitted values y_i should show only random variation, without any systematic patterns in either location or scale
- *Homoscedasticity*: A plot of the square root of the absolute values of the residuals e_i against the fitted values y_i focuses on possible changes in scale and we should see only random variation if the assumption of constant variance is appropriate
- *Normality*: A Q-Q plot is designed to show an approximately straight line when data are normally distributed and so it can be applied to the residuals to check normality
- *Independence*: It is an assumption more difficult to check, unless there are patterns in the data plotted over time, or there is some knowledge on their collection
- *Outliers*: j -th observation to be scrutinised if Cook's distance $D_j = \frac{\sum_{i=1}^n (\hat{y}_i - \hat{y}_{i(j)})^2}{p\hat{\sigma}^2}$ is large, where $\hat{y}_{i(j)}$ is the fitted value when the fit does not contain y_j

*From Biatat, 2024, AIMS

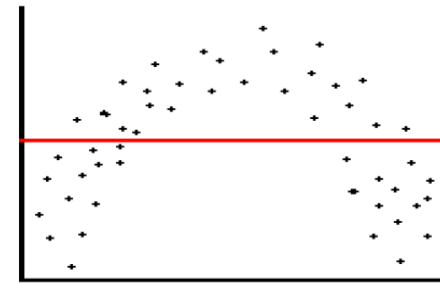
MODEL CHECKING*



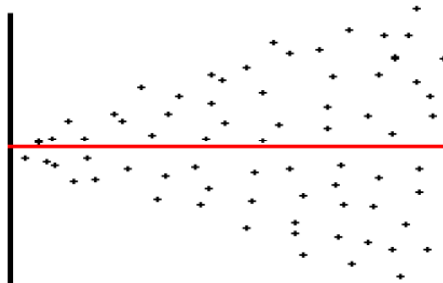
(a) Unbiased and Homoscedastic



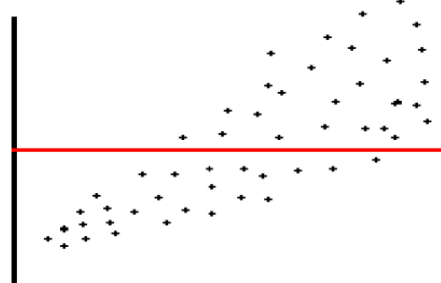
(b) Biased and Homoscedastic



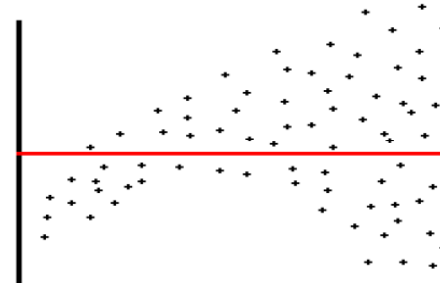
(c) Biased and Homoscedastic



(d) Unbiased and Heteroscedastic



(e) Biased and Heteroscedastic

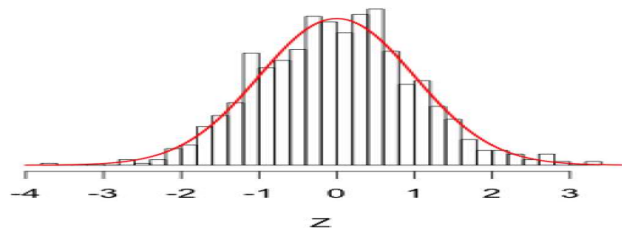


(f) Biased and Heteroscedastic

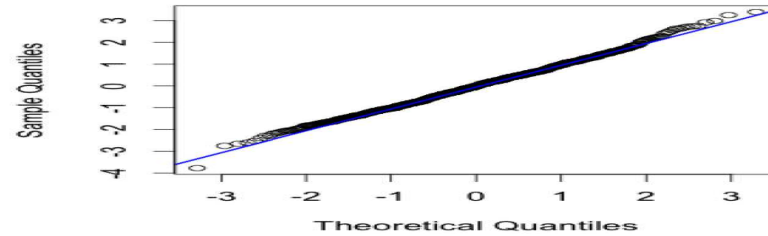
*From Biatat, 2024, AIMS

MODEL CHECKING*

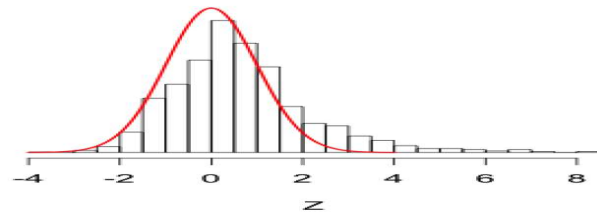
Gaussian Distribution



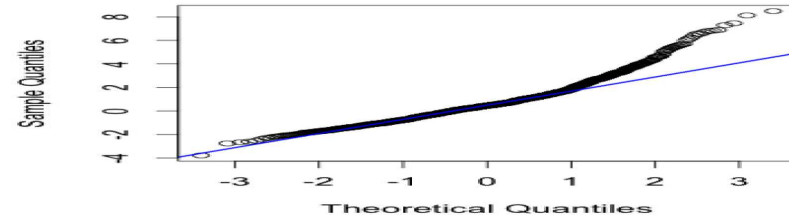
Normal Q-Q Plot



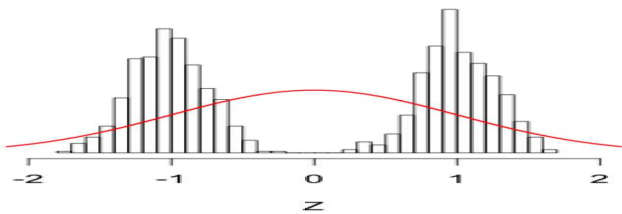
Skewed Right



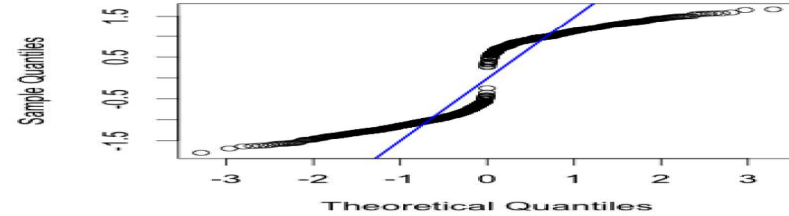
Normal Q-Q Plot



Bimodal



Normal Q-Q Plot



*From Biatat, 2024, AIMS

MODEL SELECTION

- One of the goals of a statistical analysis is to find a model suitable for the problem under investigation (like which covariates are involved and how they have an effect on a response) but, at the same time, there is a search for a parsimonious model
- In regression R^2 is used as a measure of the goodness of fit and its value should keep growing (at least, not decreasing) by adding new covariates but their contribution could be minimal w.r.t. the increase in model complexity
- For this reason, R^2 is "adjusted", penalising for the number of parameters
- This is not the unique index trying to balance better fit and parsimony: the most famous is AIC (Akaike Information Criterion), presented later
- We define the Adjusted R^2 as $R_{adj}^2 = 1 - \frac{SS_{Res}/(n - p - 1)}{SS_T/(n - 1)}$
- $\Rightarrow R_{adj}^2 = 1 - (1 - R^2)(n - 1)/(n - p - 1)$

MODEL SELECTION

- Models can differ by the number of covariates and which are in them
- Model selection is meant to search for the model which better fits the data and/or is more useful for prediction (Remember: a polynomial through all the points fits very well but it is not so good for predictions)
- We are interested in searching a regression model with a small residual sum of squares SS_{Res} but we are keen to ignore covariates which are inducing a very small decrease of SS_{Res}
- R^2_{adj} can be used to choose between models, selecting the one with the largest value
- We could use two opposite procedures to look for the covariates to be included in the models: *forward selection* which starts from the intercept alone and adds covariates one at the time and *backward elimination* which starts from all the covariates and eliminates one at the time

MODEL SELECTION

- Forward selection:
 - One starts only with the intercept β_0 and looks for the covariate, say X_1 , with the largest correlation with the response Y
 - An F test is performed to check if the covariate is significant, stopping the search for covariates if not significant
 - If X_1 is significant, then look for the covariate with the largest partial correlation, i.e. the correlation between the residuals of $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1$ and the residuals of the regression of the other candidates on X_1 , say $\hat{X}_j = \hat{\alpha}_{0j} + \hat{\alpha}_{1j} X_1$, and then perform the F test for significance
 - The procedure is repeated until either a test is not significant or there are no more covariates to add

MODEL SELECTION

- Backward elimination
 - Starts from all the covariates
 - Choose a threshold for the value of the F statistics for the partial correlation to be computed as earlier
 - Compute the partial correlation for all covariates as if they were the last one to enter in the model in the forward procedure and remove the smallest if the corresponding F statistics is below the threshold
 - Repeat the previous procedure until either an F statistics is larger than the threshold or all the covariates have been removed (and just the intercept β_0 is left)
- A stepwise regression method can be used to select covariates, adding or deleting one at the time, being a combination of the two procedures of forward selection and backward elimination, but with some adjustments

MODEL SELECTION

- Akaike Information Criterion (AIC) is a very well used index to balance goodness of fit and parsimony (also overfitting and underfitting)
- $AIC = -2 \log(\hat{L}) + 2p$, where $\hat{L}$ is the maximum of the likelihood function and p is the number of the parameters in the model
- We choose the model with the minimum AIC
- We consider the data set `mtcars` where `mpg` possibly depends on 10 covariates and we look for the model with the smallest AIC
- Choosing `both` as direction, we allow for both forward selection and backward elimination, while the alternatives are `forward` and `backward`

```
data(mtcars)
# Fit a full linear model
reg=lm(mpg ~ ., data = mtcars)
# Perform stepwise selection in both directions using AIC
reg_model=step(reg, direction = "both")
# View the summary of the final model
summary(reg_model)
```

TRANSFORMATION OF VARIABLES

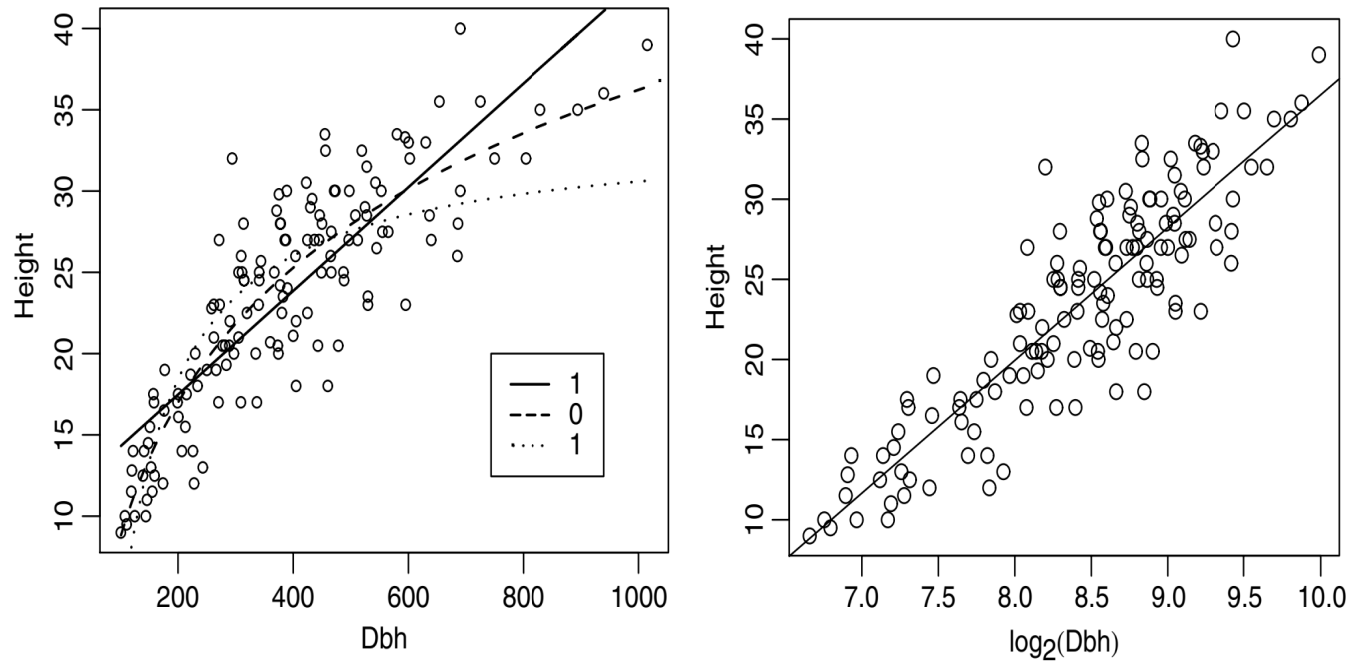
- Transforming covariates and/or response variable to use linear regression
- Dependence of tree `Height` (decimeter) on `Dbh`, diameter (mm) of the tree at 137 cm above ground, for a sample of western cedar trees collected in 1991 in the University of Idaho Experimental Forest*
- Family of scaled power transformations, defined for strictly positive x by

$$\psi(x, \lambda) = \begin{cases} (x^\lambda - 1)/\lambda & \text{if } \lambda \neq 0 \\ \log(x) & \text{if } \lambda = 0 \end{cases}$$

- Transform `Dbh` with $\psi(x, \lambda)$, $\lambda = -1, 0, 1$
- Fit `Height` with the transformed covariate
- Get back to the original scale `Dbh` with the inverse transformation
- Plot the three curves $\Rightarrow \lambda = 0$, i.e. the logarithmic transformation, looks the best

*From Weisberg, S. (2005) *Applied Linear Regression (3rd Edition)*. Wiley

TRANSFORMATION OF VARIABLES



Fitted curves ($\lambda = -1, 0, 1$) for actual (left) and transformed (logarithm) data (right)

MULTIPLE LINEAR REGRESSION

- Both frequentist and Bayesian methods will be applied in the next example
- 713 observations corresponding to the days where the prices of the Bitcoins in 8 different exchange markets were recorded together with the prices of the classical assets and the exchange rates
- We will use the package `rstanarm` and the function `stan_glm`, whose usage is similar to `lm`
- Use of improper priors leading to results close to frequentist ones
- You could try other priors, using the R tutorials, like `?stan_glm`
- For this example, I tried `stan_lm`, the very equivalent of `lm` (both about linear models) but it did not work, so that I used the one for generalised linear models
- I first present the commands for the frequentist analysis

MULTIPLE LINEAR REGRESSION

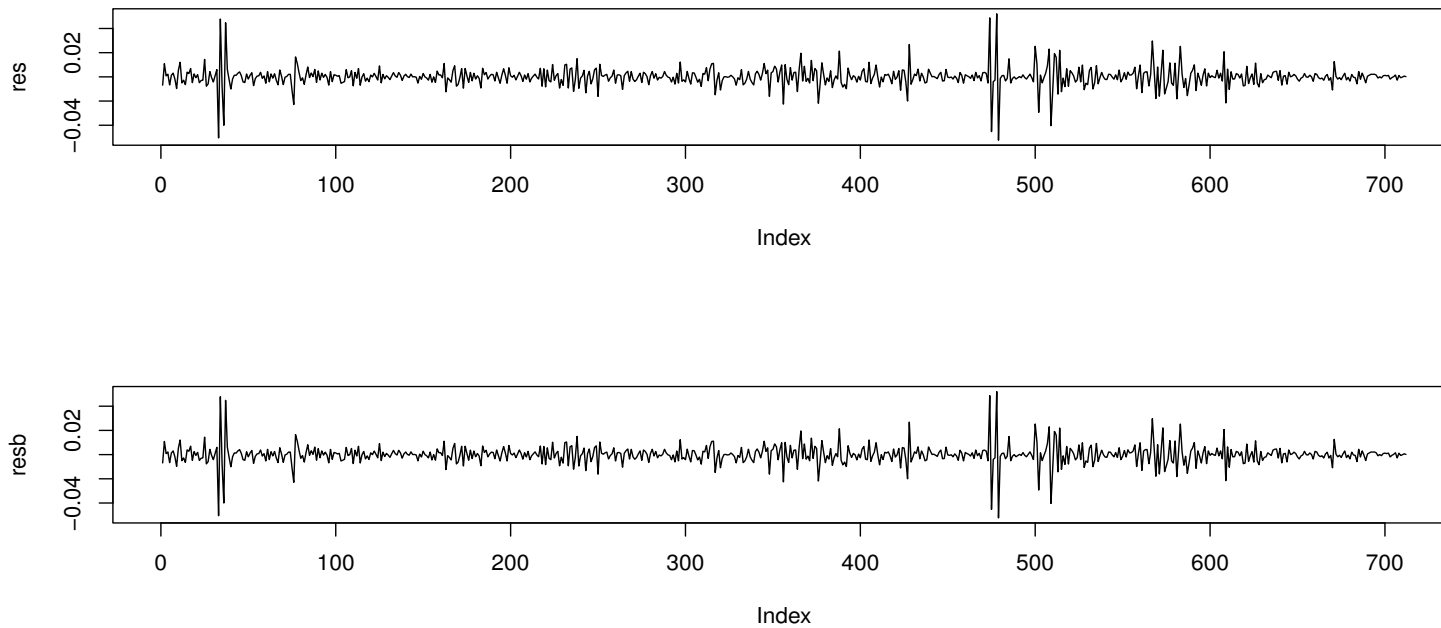
```
rm(list=ls()) # Clear the environment
install.packages("ggplot2",dependencies=TRUE)
install.packages("readxl",dependencies=TRUE)
install.packages("corrplot",dependencies=TRUE)
library(ggplot2);library(readxl);library(corrplot)
exchanges<-read_excel("exchanges.xlsx") # Read in working directory
data<-exchanges
data1<-data[-1] # Remove the first column from data
# New dataset with returns instead of prices: (log(x)-log(x-1))
data2<-as.data.frame(sapply(data1,function(x)diff(log(x),lag=1)))
attach(data2) # Bring the names of the variables directly into memory
```

MULTIPLE LINEAR REGRESSION

```
# Multiple linear regression [btc_coinbase on all other variables]
model_3<-lm(btc_coinbase~.,data=data2)
summary(model_3)
# Get and plot residuals
res<-model_3$residuals
plot(res,type='l')
install.packages("rstanarm",dependencies=TRUE)
library(rstanarm)
model_b<-stan_glm(btc_coinbase~.,data=data2)
summary(model_b, digits=3)
# Get and plot residuals
resb<-model_b$residuals
plot(resb,type='l')
```

Default priors: standard Gaussian for intercept and coefficients and exponential of parameter 1 for σ^2

MULTIPLE LINEAR REGRESSION



Residuals for MLE (top) and Bayes (bottom)

MULTIPLE LINEAR REGRESSION

Results based on MLE

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	0.0001059	0.0003123	0.339	0.73454	
btc_kraken	0.0210321	0.0123977	1.696	0.09025	.
btc_bitstamp	0.0384385	0.0359272	1.070	0.28503	
btc_itbit	0.0130343	0.0256007	0.509	0.61082	
btc_bitfinex	0.2297741	0.0315236	7.289	8.47e-13	***
btc_hitbtc	0.0821093	0.0184755	4.444	1.03e-05	***
btc_gemini	0.5981632	0.0308680	19.378	< 2e-16	***
btc_bittrex	0.0056419	0.0145595	0.388	0.69850	
usdyuan	-0.1045943	0.2066436	-0.506	0.61291	
usdeur	0.2060414	0.0986501	2.089	0.03710	*
gold	0.0712161	0.0575053	1.238	0.21597	
oil	-0.0595675	0.0192726	-3.091	0.00208	**
sp500	-0.0952889	0.0569865	-1.672	0.09495	.

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

MULTIPLE LINEAR REGRESSION

Results based on Bayes

Estimates:

	mean	sd	10%	50%	90%
(Intercept)	0.000	0.000	0.000	0.000	0.001
btc_kraken	0.021	0.013	0.005	0.021	0.038
btc_bitstamp	0.039	0.036	-0.007	0.039	0.085
btc_itbit	0.013	0.025	-0.020	0.013	0.045
btc_bitfinex	0.230	0.032	0.190	0.230	0.271
btc_hitbtc	0.083	0.019	0.059	0.083	0.106
btc_gemini	0.598	0.032	0.558	0.598	0.638
btc_bittrex	0.005	0.015	-0.013	0.005	0.024
usdyuan	-0.110	0.207	-0.373	-0.111	0.158
usdeur	0.204	0.098	0.079	0.203	0.332
gold	0.070	0.057	-0.003	0.071	0.145
oil	-0.059	0.020	-0.084	-0.060	-0.034
sp500	-0.095	0.057	-0.168	-0.096	-0.023
sigma	0.008	0.000	0.008	0.008	0.009

MULTIPLE LINEAR REGRESSION

- Warmup is better known as *burn-in*, i.e. the first values are discarded because mtcars affected by the starting values
- We now consider different priors, like Student t for each coefficient, Cauchy for the intercept and exponential for σ^2
- We consider also 1 chains, setting a seed and the number of iterations

```
model_b<-stan_glm(btc_coinbase~.,chains=1,seed=12345,iter=250,
prior=student_t(df=4,0,2.5),prior_intercept=cauchy(0,10),prior_aux =
exponential(1/2),data=data2)
summary(model_b, digits=3)
print(model_b)
prior_summary(model_b) # To see the chosen priors
library(bayesplot)
mcmc_dens(model_b)
library(bayestestR)
hdi(model_b)
```

MULTIPLE LINEAR REGRESSION

